EX. No. 3.1. CREATE 3D MODEL FROM ORTHOGRAPHIC PROJECTION

AIM:

To create a 3D model from given orthographic projection.

REQUIREMENTS:

- *Autodesk fusion 360.*
- *Windows 10*
- *Computer or Laptop*
- *Mouse or keyboard*

PROCEDURE:

1. Open Autodesk Fusion 360 and create a new design.

2. Select the XY Plane and start a new sketch.

3. Draw the main front-view profile using the given dimensions. Draw the rounded end with a radius of R16 mm and maintain the body height as 32 mm.

4. Set the distance from the hole Centre to the shoulder as 36 mm and the overall body length as 52 mm.

5. Finish the sketch and use the Extrude command to give a width of 40 mm.

6. Select the top face and create a rectangular slot at the front end. Set the slot width as 18 mm, leaving 11 mm thickness on both sides.

7. Set the slot length as 41 mm, leaving an end thickness of 11 mm.

8. Create a circular hole of 16 mm (Ø16) at the rounded end and use the Cut command.

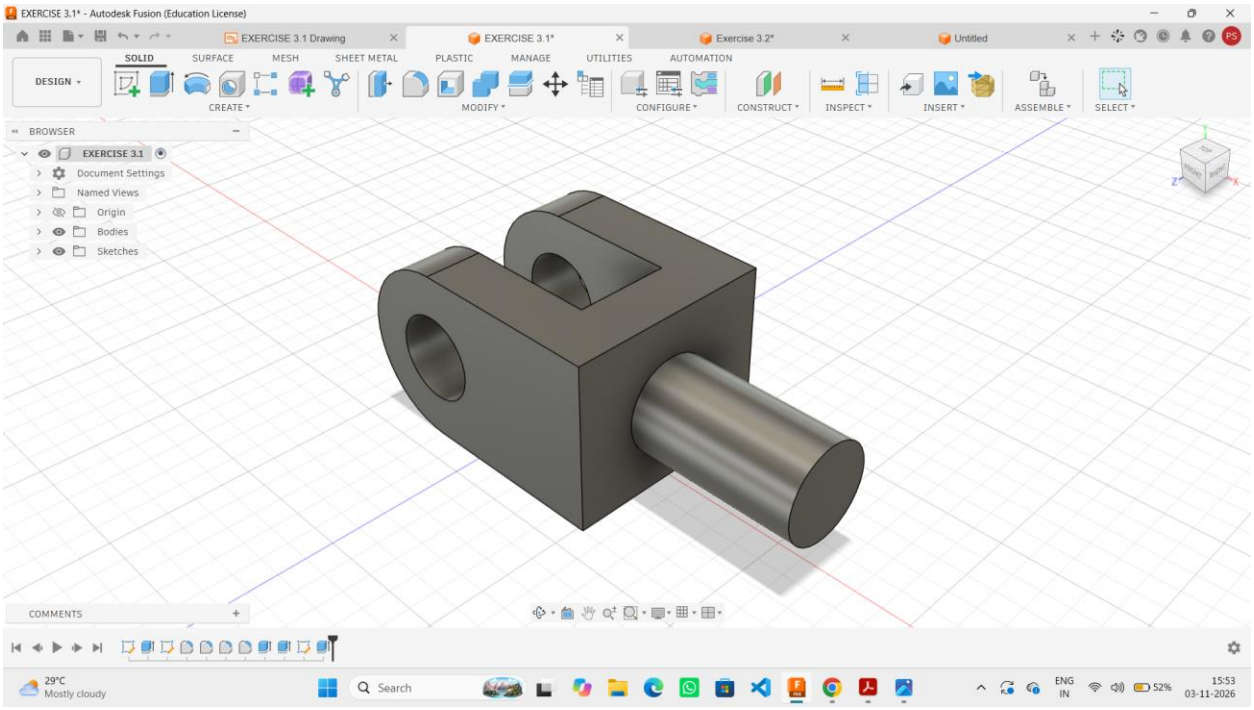
9. Select the rear face of the body and draw a circle of diameter 18 mm (Ø18) at the center.

10. Use the Extrude command to create the cylindrical portion with a length of 32 mm.

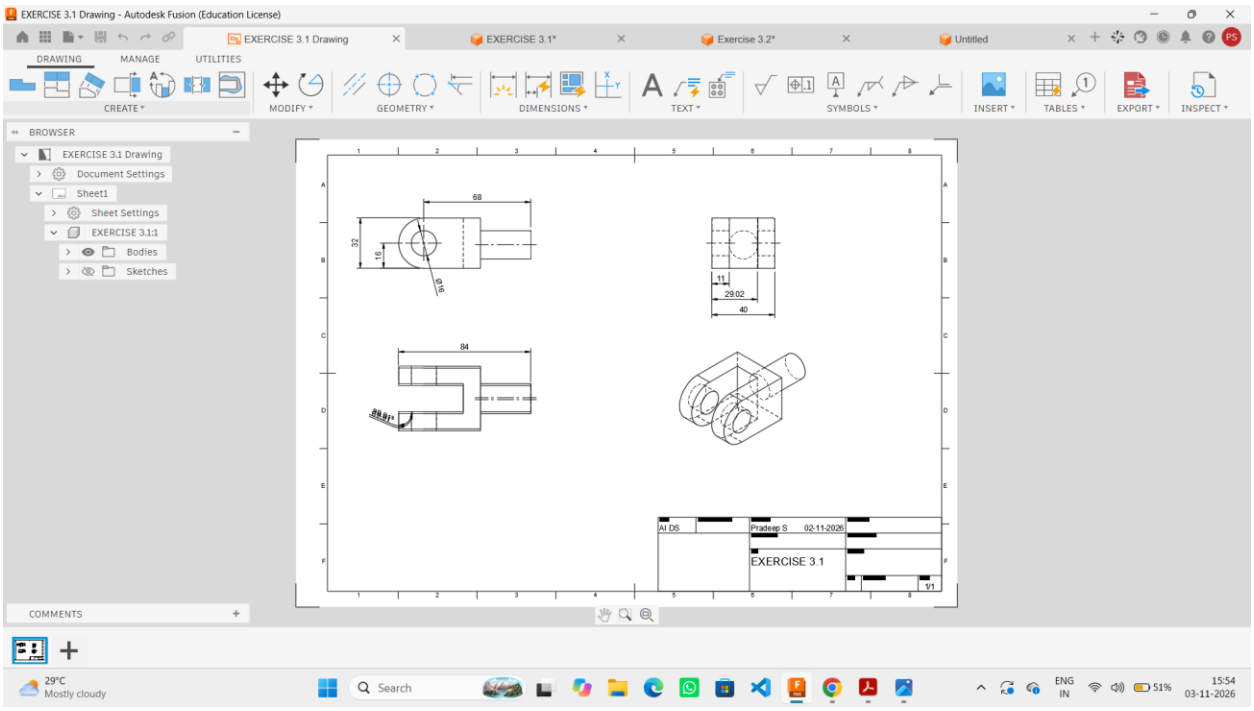
11. Apply the necessary finishing operations and check all the dimensions.

12. Finally, save the completed 3D model.

SKETCHES



OUTPUT



RESULT

Thus, 3D models are created from orthographic projections.